

29.11.2023.

Berilgan sonni tublikka tekshirish usullari.

Yevklid teoremasi: Tub sonlar toplami cheksiz.

1) 1845-yil Fransuz matematigi

J.L. Bertran:

$2a > 7$ bo'lganda

$(a; 2a-2)$ sonlar orasida kamida 1 ta tub son yotadi.

2) Natural sonlar qatorida shunday p va $p+2$ sonlar topiladiki, ularning ikkisi ham tub son bo'ladi. Bu sonlar egizak sonlar deb ataladi.

3) $x \in \mathbb{N}$ da faqat tub sonlar iborat funksiyalar mavjudmi?

$f(x) = ?$ Eyler (1707-1783)

Eyler fikri bo'yicha $x \in [1, 15]$ oraligida-
gi butun sonlar uchun $f(x) = x^2 + x + 17$

har daim tub bo'ladi.

Agar $x \in [0, 40]$ oraligida butun sonlar bo'lsa $f(x) = x^2 - x + 41$ funksiya tub bo'ladi.

4) K. Goldbax (matem) quyidagi mulohazani ilgari surgan:

Agar $x \in \mathbb{N}$ bo'lsa, barcha qiymati tub sonlardan iborat kichik $f(x)$ funksiya mavjud emas!

5) 1851-yilda rus matematigi Chebishev:

$$a \cdot \frac{x}{\ln x} < \pi(x) < A \cdot \frac{x}{\ln x}$$

$\pi(x) - x$ dan oshmaydigan tub son.

$$a > \frac{1}{2} \ln 2, \quad A < 2 \cdot \ln 2$$

Maresus ko'rinishdagi tub sonlar.

Ta'rif: Quyidagi ko'rinishdagi sonlar Ferma sonlari deb ataladi:

$$F_k = 2^{2^k} + 1, \quad k = 0, 1, 2, \dots$$

Teorema: $n = F_k$ soni $k > 0$ bo'lganda tub bōladi faqat va faqat,

$$3^{\frac{n-1}{2}} \equiv -1 \pmod{n}$$

birinchi bōlsa.

Tarif: Quyidagi kōrinishdagi sonlar **Mersen sonlari** d-di:

$$M_p = 2^p - 1, \text{ (bu yerda } p\text{-tub son)}$$

M_p tub bōlsa.

06.12.2023

Mersen sonlarining tubligini aniqlash.

Teorema: Bīror p -tub sonda ($p > 2$)

$$M_p = 2^p - 1 \text{ Mersen soni bōlishi u-n,}$$

quyidagicha ketma-ketlikda:

$$L_0 = 4, L_{j+1} = (L_j^2 - 2) \pmod{M_p},$$

$$j = 0, 1, 2, \dots, p-3,$$

ushbu

$$L_{p-1} \equiv 0 \pmod{M_p}$$

shartning bajarilishi zarur va yetarli.

Misol: $p = 5$ $M_p = 2^5 - 1 = 2^5 - 1 = 31$.

$$j = 0, 1, 2.$$

$$L_0 = 4$$

$$L_1 = (L_0^2 - 2) \pmod{31} = 14 \pmod{31} = 14.$$

$$L_2 = (L_1^2 - 2) \pmod{31} = 194 \pmod{31} = 8$$

$$L_3 = (L_2^2 - 2) \pmod{31} = 62 \pmod{31} = 0. \quad L_3 = 0$$

Demak M_5 - tub son.

Vazifa:

$$p = 13 \quad M_p = 2^{13} - 1$$

tubmi yoki tub emasmi?

Umumiy holda sonning tubligini aniqlash usullari.

Teorema (Leibniz): Agar n - natural son uchun shunday, a - butun son mavjud bōlsin:

$$a^{n-1} \equiv 1 \pmod{n} \text{ va}$$

$$a^{n-1/p} \not\equiv 1 \pmod{n}$$

$n-1$ ring bōluvchisi bōlgan p -tub sonlari uchun bajarilsa, u holda n -tub son bōladi.

Misol: $n=13$, $n-1=12$.

$p=2, 3$,

$a=2$.

$$a^{13-1} \equiv 2^{12} \pmod{13} = 16^3 \pmod{13} = 1.$$

$$2^{(13-1)/2} \equiv 2^6 \pmod{13} = 12 \not\equiv \pm 1 \pmod{13}.$$

Demak n tub son.

$$2^{(13-1)/3} = 2^4 = 16 \pmod{13} = 3.$$

Demak n -tub son.

Varifa.

$q < 13$ dagi barcha p lar topilsin.

Ma'ru:

ГОСТ Р 34.10-94 Rossiya Fed. ERI

standarti (1994-2000) algoritmi uchun

tub sonlarni generatsiya qilish

usuli.

ГОСТ - SSSR standarti.

ГОСТ Р - Rossiya standarti.

ECDSA - AQSh ning 2000-yildan boshlab
ERI standarti (elliptik epi chizigga asoslan)

ГОСТ Р 34.10-2001 $\Rightarrow$ RF ning ERI standarti.

Teorema: q - toq tub son, $p = q \cdot N + 1$,

bu yerda N - juft son bolsin. Shuningda

$p < (2q+1)^2$ va quyidagi shartlar:

1) $2^{q \cdot N} \equiv 1 \pmod{N}$

2) $2^N \not\equiv 1 \pmod{p}$

bajarilsa, u holda p -tub son.

Varifa.

$p=61$, $q=5$, $N=12$

Yechish:

1) $p=13$ $M_p = 2^{13} - 1$ tub sonmi?

$L_0 = 4$

$j = 0, 1, \dots, 10$, $M = 2^{13} - 1 = 8191$.

$L_1 = (L_0^2 - 2) \pmod{8191} = 14$.

$L_2 = (L_1^2 - 2) \pmod{8191} = 194$

$L_3 = (L_2^2 - 2) \pmod{8191} = 4870$

$L_4 = (L_3^2 - 2) \pmod{8191} = 3953$

$L_5 = (L_4^2 - 2) \pmod{8191} = 5970$

$L_6 = (L_5^2 - 2) \pmod{8191} = 1857$

$L_7 = (L_6^2 - 2) \pmod{8191} = 36$

$L_8 = (L_7^2 - 2) \pmod{8191} = 1294$

$L_9 = (L_8^2 - 2) \pmod{8191} = 3470$

$$L_{10} = (4^2 - 2) \pmod{8191} = 128$$

$$L_{11} = (4^2 - 2) \pmod{8191} = 0$$

Demak 8191-tub son.

2) $n=13$, $n-1=12$, $p=2, 3$, $a \in [2, 11] \in \mathbb{N}$.

$$a=2: a^{13-1} = a^{12} = 2^{12} \pmod{13} = 16^3 \pmod{13} = 3^3 \pmod{13} = 1.$$

$$p=2: 2^{\frac{12}{2}} = 2^6 = 64 \pmod{13} = 12 \neq 1.$$

$$p=3: 2^{\frac{12}{3}} = 2^4 = 16 \pmod{13} = 3 \neq 1$$

$$\times a=3: 3^{12} \pmod{13} = 27^4 \pmod{13} = 1.$$

$$p=2: 3^{\frac{12}{2}} = 3^6 \pmod{13} = 27^2 \pmod{13} = 1.$$

$$\times a=4: 4^{12} \pmod{13} = 16^6 \pmod{13} = 3^6 \pmod{13} = 1.$$

$$p=2: 4^{\frac{12}{2}} = 4^6 \pmod{13} = 16^3 \pmod{13} = 3^3 \pmod{13} = 1.$$

$$\times a=5: 5^{12} \pmod{13} = 25^6 \pmod{13} = (-1)^6 \pmod{13} = 1$$

$$p=2: 5^6 \pmod{13} = 25^3 = (-1)^3 = -1 = 12 \neq 1.$$

$$p=3: 5^4 \pmod{13} = 25^2 = (-1)^2 = 1$$

$$a=6: 6^{12} \pmod{13} = 36^6 = (-3)^6 = 3^6 = 27^2 = 1$$

$$p=2: 6^6 \pmod{13} = 36^3 = (-3)^3 = -27 = 12 \neq 1.$$

$$p=3: 6^4 \pmod{13} = 36^2 \pmod{13} = (-3)^2 = 9 \neq 1$$

$$a=7: 7^{12} \pmod{13} = 49^6 = (-3)^6 = 27^2 = 1$$

$$p=2: 7^6 \pmod{13} = 49^3 \pmod{13} = (-3)^3 = -27 = 12 \neq 1$$

$$p=3: 7^4 \pmod{13} = 49^2 = (-3)^2 = 9 \neq 1$$

$$\times a=8: 8^{12} \pmod{13} = 64^6 = 12^6 = (-1)^6 = 1.$$

$$p=2: 8^6 \pmod{13} = 64^3 = 12^3 = (-1)^3 = -1 = 12 \neq 1.$$

$$p=3: 8^4 \pmod{13} = 64^2 = 12^2 = (-1)^2 = 1$$

$$\times a=9: 9^{12} \pmod{13} = 81^6 \pmod{13} = 3^6 = 27^2 = 1.$$

$$p=2: 9^6 \pmod{13} = 81^3 = 3^3 = 27 = 1$$

$$\times a=10: 10^{12} \pmod{13} = 100^6 = 9^6 = 81^3 = 3^3 = 1.$$

$$p=2: 10^6 \pmod{13} = 100^3 = 9^3 = 81 \cdot 9 = 3 \cdot 9 = 27 = 1$$

$$a=11: 11^{12} \pmod{13} = (-2)^{12} = 2^{12} = 16^3 = 3^3 = 27 = 1.$$

$$p=2: 11^6 = (-2)^6 = 2^6 = 64 = 12 \neq 1.$$

$$p=3: 11^4 = (-2)^4 = 2^4 = 16 = 3 \neq 1$$

Jawab: $a = 2, 6, 7, 11$.

3) $p=61$, $q=5$, $N=12$.

1) $2^{q \cdot N} \equiv 1 \pmod{p}$

$$2^{5 \cdot 12} \equiv 2^{60} \pmod{61} = 1 \quad 1\text{-short bajarildi.}$$

2) $2^N \not\equiv 1 \pmod{p}$

$$2^{12} \not\equiv 1 \pmod{61} = 9 \neq 1 \quad 2\text{-short bajarildi.}$$

demak $p (p=61)$ soni-tub son.

13.12.2023.

Primitiv ildiz

Ta'rif: Aytaylik p -tub son va $q < p$ bolsin. U holda q soni p modul boyicha primitiv ildiz (p modul boyicha tashkil etuvchi) deyiladi, agar har bir b soni ($1 < b < p-1$) uchun shunday a mavjud bolsaki, $q^a \equiv b \pmod{p}$ bo'lsin.

p -tub son.

$q_1, q_2, \dots, q_n - (p-1)$ sonining tub bo'luvchilari.

q soni p modul boyicha primitiv ildiz bolishi uchun $q^{\frac{p-1}{q_i}} \pmod{p} \neq 1$

bolishi lozim. Aks holda, ya'ni biron

q_i da $q^{\frac{p-1}{q_i}} \pmod{p} = 1$ bolsa?

$n=13$ boyicha olingan natijalar

(2, 6, 7, 11) primitiv ildiz ekan.

(3, 4, 5, 8, 9, 10) lar primitiv ildiz emas ekan.

Berilgan sonning tubligini

aniqlashning ayrim usullari

Teorema (Pohligton). Aytaylik $n \geq 3$

son va $n = Q \cdot R + 1$, bu yerda

$\text{EKUB}(Q, R) = 1$ va $Q = \prod_{d=1}^k q_i^{d_i} \rightarrow Q$ ning tub bo'luvchilarga yoyilma

agar har bir q_i -uchun shunday

a_j mavjud bolsa va $a_j^{n-1} \equiv 1 \pmod{n}$ har

$\text{EKUB}(a_j^{\frac{n-1}{q_i}} - 1, n) = 1$ bolsa, u holda

n -tub son boladi.

Vazifa

Pohligtonga misol

Yechish.

$n=7$ $Q=6$ $R=1$.

$n = Q \cdot R + 1 \rightarrow 7 = 6 \cdot 1 + 1$

$\text{EKUB}(Q, R) = (6, 1) = 1$.

$Q = 2 \cdot 3$ $q_1 = 2$, $q_2 = 3$.

$a = 3$

① $a^{n-1} \equiv 1 \pmod{n}$.

$3^6 \pmod{7} = 3^3 \pmod{7} = 2^3 \pmod{7} = 1$.

$\text{EKUB}(3^{\frac{6}{2}} - 1, 7) = (26, 7) = 1$.

② $\text{EKUB}(3^{\frac{6}{3}} - 1, 7) = (8, 7) = 1$.

Demak $n=7$ tub son.

22.12.2023

Berilgan sonni tublikka tekshiruvchi ehtimollik testlari
(Solovay - Shtraassen va Miller-Rabin testlari)

Berilgan natural toq n soni tub ekanligini aniqlashning ehtimollik testi quyidagi tartibda amalga oshiriladi.

$\exists a \in \mathbb{N}$, $1 < a < n$ soni tanlanadi, va uning uchun ma'lum bir short-larning bajarilishi tekshiriladi. Agar bu shartlarning birozasi bajarilmasa n murakkab son bo'ladi.

Agar qaysidir a qiymatda keltirilgan barcha shartlar bajarilsa u holda n soni tub degan xulosa qilinib qolmaydi. Biroq n soni ma'lum bir ehtimollik bilan tub deyiladi.

Teorema: Aytaylik, n -toq murakkab son. U holda quyidagi shartlarni:

1) $\text{EKUB}(a, n) = 1$

2) $a^{\frac{n-1}{2}} \equiv \pm 1 \pmod{n}$

qanoatlantiruvchi $1 < a < n$ oralig'dagi olingan butun a -lar soni $n/2$ dan oshmaydi.

Bu yerda $f\left(\frac{a}{n}\right) \rightarrow$ yakabi simvoli.

Solovay - Shtraassen testi

1) $1 < a \leq n-1$ tasodifiy k ta a sonini olamiz, ($k < n$)

2) tasodifiy olingan k ta a larga nisbatan teoremaning 1-va 2-shartlarini tekshiramiz. Birinchi bolsa 3-qadamga o'tamiz.

3) U holda, n -soni $\left(1 - \frac{1}{2^k}\right) = p$ ehtimollik bilan n soni tub son deb e'lon qilinadi.

Miller - Rabin ehtimollik testi.

Aytaylik n -toq murakkab son va 3 ga bo'linmasin,

$n-1 = 2^r \cdot t$ ifoda o'rinli bolsin, bu yerda $r \geq 1$, t -toq son. U holda

$$a^t \equiv 1 \pmod{n}$$

yoki

$$a^{\frac{n-1}{2}} \equiv -1 \pmod{n},$$

ushbu, $1 \leq j \leq r$ olalgidan olingan biror j uchun, shartlarni qanoatlantiruvchi $0 < a \leq n-1$ oralig'idan olga olingan a -lar soni $n/4$ dan oshmaydi.

Natija: Teorema dan agar k ta tasodifiy a ning qiymatlari uchun n sonining murakkab emasligi aniqlansa, u holda berilgan n soni $(1 - \frac{1}{4^k})$ ehtimollik bilan tub son deb quritiladi.

27.12.2023

Butun sonni tub ko'paytuvchilarga ajratishning zamonaviy usullari tahlili.

Tarif: Agar n mustat butun son uchun quyidagi kanonik $n = p_1^{a_1} \cdot p_2^{a_2} \cdot \dots \cdot p_s^{a_s}$ yozilmani topish masalasi qarayotgan bo'lsa, bu holda bunday masalaga berib, n murakkab sonni tub ko'paytuvchilarga ajratish (yoki faktORIZATSIYA) deyiladi.

Amalda, xususan kriptologiya fanida $n = p \cdot q$ sonni tub ko'paytuvchilarga ajratish masalasi qaratiladi.

Bugungi kunda faktORIZATSIYA masalalarida arisii ilqarilash bosqichlari si-fatida chiziqli reshetta kvadratik reshetta va sonli maydon reshetta usullari alohida ajratib krsatiladi.

Ferma usuli

Teorema: Aytaylik $n > 1$ toq son. Bu son murakkab son bo'ladi. Faqat va faqat $p|q \in \mathbb{Z}^+$ bo'lib, $n = p^2 \cdot q^2$ bo'lsa, Bu yerda $p, q > 1$.

Ferma usulining mohiyati shundan iboratki, teorema natijasiga ko'ra $\exists p, q \in \mathbb{Z}^+$ sonlar topish kerakki, $n = p^2 \cdot q^2$; $p^2 = n + q^2$, yoki $q^2 = n - p^2$ bajarilsin. Agar $p^2 = n + q^2$, $q = 1, 2, 3, \dots$ qiymatlar uchun $n + q^2$ biror sonning to'la kvadratlidan iborat bo'lmasa, n holda $q = (n-1)/2$ qiymat uchun $n + q^2$ ni tekshirib ko'ramiz va biror sonning kvadratlidan iborat bo'lsa. Aks holda n -tub son bo'ladi.

Masalan: $n = 527$.

q	$n + q^2$
1	$527 + 1 = 528$
2	$527 + 4 = 531$
3	$527 + 9 = 536$
4	$527 + 16 = 543$
5	$527 + 25 = 552$
6	$527 + 36 = 563$
7	$527 + 49 = 576 = 24^2$

Varika

$$n=3 \quad q_{\max} = (n-1)/2 = (3-1)/2 = 1$$

$$q=1 \Rightarrow p^2 = n + q^2 = 3 + 1 = 4 \Rightarrow p=2 \quad n = (p-q)(p+q) = 1 \cdot 3$$